Earth's rotation. Similarly, all of Gargantua's primary stellar images circulate around the red pole-star images as the camera orbits the hole, but their circulation paths (for example, the two red eddy curves) are highly distorted by the whirl of space and gravitational lensing. Similarly, all the secondary stellar images circulate around the yellow pole-star images (for example, along the two distorted yellow curves).

Why, for a nonspinning hole (Figure 8.4), did the secondary images appear to emerge from the black hole's shadow, swing around the hole, and descend back into the shadow, instead of circulating around a closed curve as for Gargantua (Figure 8.5)? They actually *do* circulate around closed curves for a nonspinning hole. However, the inner edge of the closed curve is so close to the shadow's edge that it can't be seen. Gargantua's spin makes space whirl, and that whirl moves the inner Einstein ring outward, revealing the secondary images' full circulatory pattern (yellow curves in Figure 8.5), and revealing the inner Einstein ring.

Inside the inner Einstein ring, the streaming pattern is more complicated. The stars in this region are tertiary and higher-order images of all the stars in the universe—the same stars as appear as primary images outside the outer Einstein ring and secondary images between the Einstein rings.

In Figure 8.6, I show five small pictures of Gargantua's equatorial plane, with Gargantua itself in black, the camera's orbit in dashed purple, and a light ray in red. The light ray brings to the camera the stellar image that is at the tip of the blue arrow. The camera is moving counterclockwise around Gargantua.

You can get a lot of insight into the gravitational lensing by walking yourself through these pictures, one by one. Take note: The actual direction to the star is upward and rightward (see outer ends of the red rays). The camera and beginning of each ray point toward the stellar